Balancing Security and Civil Liberties: A Review of AI Surveillance and Facial Recognition Technology

Abstract

The rapid integration of Artificial Intelligence (AI) and Face Recognition Technologies (FRT) into surveillance systems has created serious tension between public security and fundamental civil liberties. This review combines recent research to analyze the key ethical challenges and policy debates arising from technological shift. One key problem identified in the literature is a governance crisis. Surveillance technologies are expanding. However, laws and regulations are not developing fast enough to control their growth. Current policy responses, such as using data protection laws like EU's GDPR, are explored but shown to be reactive and inconsistent, addressing problems after the harm. Collectively, the review concludes that navigating this complex landscape requires a proactive, multi-stakeholder governance model.

Keywords: AI surveillance, Facial Recognition Technology, Civil Liberties, Privacy, Algorithmic Bias, Regulation

Introduction

According to (Choung, et al., 2024), AI-powered surveillance uses machine learning and large amounts of data from daily activities, such as cameras and online images, to monitor, track and analyze human behavior in real time. The authors further note that many countries are now heavily investing in Facial Recognition Technology (FRT) as part of AI surveillance systems. FRT identifies individuals based on their unique facial features, enabling both real-time and post-hoc identification.

Likewise, the authors highlight ethical concerns surrounding AI-powered surveillance, particularly the loss of personal privacy and threat to civil liberties. As AI technologies become more advanced, they enable large-scale or mass surveillance, raising serious concerns over data access, control, usage and long-term storage of personal information.AI surveillance systems may contain algorithmic bias, leading to unfair treatment of certain groups. There is a risk of discrimination based on race, ethnicity or other characteristics. Surveillance data can be misused by governments or organizations for oppressive monitoring, authoritarian control or targeting dissent or vulnerable populations. (Mosa, et al., 2024)

Overall, AI-powered surveillance, especially facial recognition, creates a difficult balance between increasing security and safety, protecting privacy, human rights, and fairness.

According to (Arora, et al., 2025), while public safety must be ensured, individual rights should not be ignored. Achieving this balance requires cooperation between technologists, ethicists and lawmakers. As cited in (Arora, et al., 2025), (Wright, 2018) emphasizes the importance of open negotiation to determine where AI surveillance should be permitted and where it should be restricted. The researchers conclude that the development of such balanced regulatory frameworks would allow AI's benefits to be effectively utilized while minimizing harm to civil liberties.

Thematic Review

Research has shown that in recent years, Facial Recognition Technology (FRT), a subset of Artificial Intelligence (AI), has developed rapidly and become embedded in everyday life. Globally, many organizations use FRT for purposes such as verifying identity online identities, approving digital payments, monitoring employee attendance and targeting advertisements to consumers (Handelman, 2023).

Law enforcement agencies have been among the earliest and most active users of FRT, driven by need to detect crimes more efficiently, successfully prosecute offenders and enhance public safety. While FRT offers clear operational benefits, its deployment has significantly altered the relationship between police and citizens. The expansion of such new surveillance technologies has increased monitoring capabilities while simultaneously reducing levels of personal privacy (Almeida, et al., 2021).

Concerns surrounding the misuse of FRT became especially prominent in January 2020, when journalist Kashmir Hill exposed Clearview AI, a company that developed an exceptionally powerful facial recognition system. Clearview AI secretly scraped photographs from the internet and social media platforms, gathering a database of approximately 10 billion images worldwide by October 2021. The tool allowed users to identify individuals and access linked images from multiple online sources. Although the company claimed that its services only restricted to law enforcement agencies in the USA and Canada, investigations later revealed that private companies and individual users were also utilizing the technology, raising serious ethical and legal concerns (Dul, 2022).

Public debate surrounding FRT is further illustrated by the case of San Francisco. As discussed by (Choung, et al., 2024), the city initially banned the use of facial recognition by police and other government agencies due to concerns over privacy and civil liberties. However, in 2024, city voters approved a policy shift granting police broader authority to use surveillance technologies, including facial recognition systems and drones. This reversal highlights the ongoing tension in public opinion, where some citizens prioritize privacy and civil liberties, while others prioritize safety, crime prevention and policing efficiency.

Facial Recognition Technology (FRT), a system that identifies people by analyzing their facial features using computer algorithms, can be unfair or biased. This occurs when the data used to train algorithms is not representative. As explained in both (Mosa, et al., 2024) and (Handelman, 2023), if the datasets are predominantly composed of certain groups, especially white men, the

AI becomes proficient at recognizing them but performs poorly on others such as women or people of color.

A well-known example from the United States, detailed in (Handelman, 2023), in a case of an African American man named Robert Julian-Borchark Williams. He was wrongly arrested after a facial recognition system mistakenly matched his face to low-quality photo taken from a store's security camera. The system failed to accurately identify him, highlighting how this inherent bias can lead to serious real-world consequences.

Since AI surveillance technologies collect and analyze large amounts of personal data, particularly sensitive biometric data such as facial images, researchers emphasize the importance of regulatory safeguards to protect individual rights while still allowing these allowing technologies to be used for security purposes. Authors note that governments are increasingly attempting to establish legal frameworks that balance public safety with the protection of civil liberties.

Some regions have already implemented strong legal protections. For example, in European Union (EU), the General Data Protection Regulation (GDPR) strictly governs how personal data is collected, processed and used. Under GDPR, organizations must have a clear and lawful purpose for collecting personal data, ensure it is securely protected and respect individuals rights to be informed about how their data is used (Arora, et al., 2025) also mentioned in (Dul, 2022).

Similarly, other researchers propose additional measures to safeguard civil liberties while permitting the use of AI surveillance technology. These include the establishment of international rules and the creation of a global regulatory body to oversee AI deployment. Beyond regulation alone, researchers argue that effective governance requires close cooperation between policymakers and private sector. Policymakers are responsible for enacting robust and enforceable laws, while private companies must design and deploy AI systems in ways that are ethically responsible and respectful of privacy and civil liberties. To support this approach, global standards should be strengthened to regulate AI frameworks, promote ethical design and implementation and ensure transparency & accountability through ethical research and international collaboration (Mosa, et al., 2024).

Discussion

In conclusion, this debate surrounding the ethical use of AI in surveillance and facial recognition is no longer a theoretical dilemma but a pressing governance crisis. Although research acknowledges the significant utility of these technologies for security and efficiency, existing evidence reveals a consistent failure to establish adequate safeguards, resulting in real and measurable harm to civil liberties. One major concern identified across the literature is function creep, the gradual expansion of surveillance systems beyond its original intended purpose, which raises serious privacy and ethical issues. A clear example of this is the case of Clearview AI, where facial recognition tools were developed using large amounts of unauthorized and scraped personal data.

The research also highlights the issue of algorithmic bias within FRT, demonstrated by cases such as the wrongful arrest of Robert Julian-Borchak Williams due to false facial recognition match. These errors show that the technical flaws in FRT are closely connected to broader patterns of social injustice, particularly affecting marginalized communities. Furthermore, the literature emphasizes that effective governance cannot be achieved by any single group alone. Instead, it requires strong collaboration between policymakers and private technologies companies. While regulatory frameworks such as the EU's GDPR provides a useful foundation, the research pressures the need for stronger international standards supported by transparency, accountability and human right protection. Overall, the findings conclude that proactive and preventive governance is essential to ensure that AI surveillance technologies do not undermine civil liberties.

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